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METHOD FOR PROBE PIN RETRIEVAL

Abstract

A method for re-using a probe pin includes following operations. A first probe head including an upper die, a lower die, and at least a probe pin is received. A first bending delta between a pin tip and a pin head of the probe pin is compared with a value. The probe pin is bended to obtain a second bending delta when the first bending delta is greater than the value. The probe pin is pushed in a first direction from the lower die to the upper die after the bending of the probe pin, or when the first bending delta is less than the value. The probe pin is retrieved from the first probe head, and assembled to a second probe head. A third bending delta of the probe pin in the second probe head is less than the first bending delta.

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Background/Summary

PRIORITY DATA [0001] This application is a continuation of U.S. patent application Ser. No. 17/687,400 filed on Mar. 4, 2022, entitled of “METHOD FOR PROBE PIN RETRIEVAL”, which claims the benefit of U.S. Provisional Patent Application Ser. No. 63/227,745 filed Jul. 30, 2021, the entire disclosure of which is hereby incorporated by reference.

BACKGROUND

[0002] During various stages in semiconductor manufacturing, there is a need to verify quality of formed integrated circuits. Testing of the integrated circuits may be performed by using probe cards to probe contact pads of metal bumps on surfaces of the wafers or chips.

[0003] As pitches of the contact pads are reduced, corresponding pitches of probe pins in the corresponding probe card need to be reduced accordingly. As a result, a challenge arises in assembly of probe heads that includes such small-pitch probe pins.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIG. 1 shows a flowchart representing a method for a probe pin retrieval according to aspects of the present disclosure in one or more embodiments.

[0006] FIG. 2 is a schematic drawing illustrating a probe head according to aspects of the present disclosure in one or more embodiments.

[0007] FIG. 3 is a schematic drawing illustrating a probe pin according to aspects of the present disclosure in one or more embodiments.

[0008] FIGS. 4A and 4B are schematic drawings illustrating a probe head in different stages according to aspects of the present disclosure in one or more embodiments.

[0009] FIGS. 5 to 11 illustrate a method for a probe head pin retrieval at various stages according to aspects of the present disclosure in one or more embodiments, wherein FIG. 10 is a schematic drawing illustrating a probe pin retrieved from the probe head.

[0010] FIG. 12 shows a flowchart representing a method for a probe pin retrieval according to aspects of the present disclosure in one or more embodiments.

DETAILED DESCRIPTION

[0011] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat

reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0012] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” “on” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0013] As used herein, terms such as “first,” “second” and “third” describe various elements, components, regions, layers and/or sections, but these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another. The terms such as “first,” “second” and “third” when used herein do not imply a sequence or order unless clearly indicated by the context.

[0014] Notwithstanding that the numerical ranges and parameters setting forth the broad scope of the disclosure are approximations, the numerical values set forth in the specific examples are reported as precisely as possible. Any numerical value, however, inherently contains certain errors necessarily resulting from the normal deviation found in the respective testing measurements. Also, as used herein, the terms “substantially,” “approximately” or “about” generally mean within a value or range that can be contemplated by people having ordinary skill in the art. Alternatively, the terms “substantially,” “approximately” or “about” mean within an acceptable standard error of the mean when considered by one of ordinary skill in the art. People having ordinary skill in the art can understand that the acceptable standard error may vary according to different technologies. Other than in the operating/working examples, or unless otherwise expressly specified, all of the numerical ranges, amounts, values and percentages such as those for quantities of materials, durations of times, temperatures, operating conditions, ratios of amounts, and the likes thereof disclosed herein should be understood as modified in all instances by the terms “substantially,” “approximately” or “about.” Accordingly, unless indicated to the contrary, the numerical parameters set forth in the present disclosure and attached claims are approximations that can vary as desired. At the very least, each numerical parameter should be construed in light of the number of reported significant digits and by applying ordinary rounding techniques. Ranges can be expressed herein as being from one endpoint to another endpoint or between two endpoints. All ranges disclosed herein are inclusive of the endpoints, unless specified otherwise.

[0015] A large number of probe cards are used for device testing. Such probe cards may not be used again at the end of their service life. However, probe pins, from the probe cards may be recycled and re-used. Therefore, a method for retrieving and re-assembling the probe pins are required.

[0016] In some embodiments, a probe card is configured to test functionality of devices-under-test (DUTs). In some embodiments, the DUTs may be semiconductor devices, chips, or dies with contact pads on a wafer. In some embodiments, a probe card may include at least a circuit board (not shown), a space transformer (not shown), a probe head **200** and at least a probe pin **300**. In some embodiments, the circuit board may be a printed circuit board (PCB) having certain testing circuits. In some embodiments, the circuit board electrically connects the DUTs to an automatic testing equipment (ATE) for testing. The space transformer is disposed between the circuit board and the probe head **200** to electrically connect the probe head **200** to the circuit board.

[0017] Please refer to FIG. 2, which is a schematic drawing illustrating a probe head **200** of a probe card. The probe head **200** may include at least an upper die **210** (also referred to as an upper guide plate), at least a lower die **220** (also referred to as a lower guide plate), and a spacer **230** between the upper die **210** and the lower die **220**. In some embodiments, a quantity of the upper dies **210**

may be modified according to different requirements. For example, the probe head **200** may include a first upper die, and a second upper die between the first upper die and the spacer **230**, but the disclosure is not limited thereto. In such embodiments, a connection element may be disposed between the first upper die and the second upper die to secure the two upper dies, but the disclosure is not limited thereto. In some embodiments, a quantity of the lower dies **220** may be modified according to different requirements. For example, the probe head **200** may include a first lower die, and a second lower die between the first lower die and the spacer **230**, but the disclosure is not limited thereto. In such embodiments, a connection element may be disposed between the first lower die and the second lower die to secure the two lower dies, but the disclosure is not limited thereto. The connection elements include, for example but not limited thereto, pins, clamps, or the like.

[0018] The upper and lower dies **210** and **220** are configured to constrain the probe pin **300**. In some embodiments, the upper die **210** and the lower die **220** may be formed of a dielectric material such as ceramic, vespel, engineering plastic, or the like, but the disclosure is not limited thereto. In some embodiments, the upper die **210** and the lower die **220** may include silicon, silicon nitride, and quartz, but the disclosure is not limited thereto. An example range of thickness for each of the upper die **210** and the lower die **220** may be between approximately 250 micrometers and approximately 675 micrometers, but the disclosure is not limited thereto. The term “thickness” as related to the upper and lower dies **210** and **220** indicates a vertical dimension in the context of FIG. 2. In some embodiments, the thickness of each of the upper die **210** and the lower die **220** may be approximately 500 micrometers, but the disclosure is not limited thereto.

[0019] The spacer **230** is disposed between the upper die **210** and the lower die **220**. The spacer **230**, the upper die **210**, and the lower die **220** together define a cavity C. A probe pin **300** is disposed in the cavity C and extends from the lower die **220** toward the upper die **210**. It should be noted that the spacer **230** can be plural. In such embodiments, there may be two separate spacers **230** inserted from opposite directions toward the space between the lower die **220** and the upper die **210**. In other embodiments, there may be three or four spacers **230**. The spacers **230** may be formed of a dielectric material, although they may also be formed of conductive materials. For example, in some embodiments, the spacers **230** may include metals, such as steel, or other rigid materials that have good flatness and provide stability for the upper and lower dies **210** and **220**. After the spacers **230** are inserted into desirable positions, the relative positions of the upper die **210**, the lower die **220**, and the spacers **230** may be secured, for example, by using pins, clamps, or the like. The probe head **200** is thus assembled. An example range of a thickness for the spacer **230** may be between approximately 3 millimeters and approximately 5 millimeters. For example, the thickness of the spacer **230** is approximately 4 millimeters, but the disclosure is not limited thereto.

[0020] The upper die **210**, the lower die **220** and the spacer **230** may have a variety of shapes and dimensions, depending on a required design. For example, the upper die **210**, the lower die **220** and the spacer **230** may be rectangular or circular in shape, or may have irregular shapes.

[0021] Still referring to FIG. 2, in some embodiments, the upper die **210** has at least a through hole **214** (also referred to as an aperture) allowing the probe pin **300** to pass through. Similarly, the lower die **220** has at least a through hole **224** allowing the probe pin **300** to pass through. It should be noted that the through hole **214** of the upper die **210**, the through hole **224** of the lower die **220**, and the probe pin **300** can be plural, as shown in FIG. 2. In some embodiments, the through holes **214** of the upper die **210** and the through holes **224** of the lower die **220** are vertically aligned with a one-to-one correspondence.

[0022] The through holes **214** and **224** may be rectangular or circular in shape but may be varied in accordance with some other embodiments of the present disclosure. An example range of widths of the through holes **214** and **224** may be between approximately 50 micrometers and approximately 70 micrometers, where the term “width” as related to the through holes **214** and **224** indicates a horizontal dimension in the context of FIG. 2. The widths of the through holes **214** may be

substantially equal to each other, and the widths of the through holes **224** may be substantially equal to each other, but the disclosure is not limited thereto. In some embodiments, the widths of the through holes **214** and the widths of the through holes **224** are substantially equal to each other, but the disclosure is not limited thereto.

[0023] The through holes **214** and **224** are arranged according to a patterns of contact pads of a DUT (not shown) with a one-to-one correspondence, so that the probe head **200** may be used to probe the DUT. In some embodiments, the through holes **214** and **224** are arranged as an array, although other patterns may be used, depending on the pattern of the DUT.

[0024] Referring to FIGS. **2** and **3**, in some embodiments, the probe pin **300** is formed of conductive materials such as metals. In some embodiments, the probe pin **300** may have a circular cross-section. In other embodiments, the probe pin **300** may have a rectangular (such as a square) cross-section (a top view or a bottom view), but the disclosure is not limited thereto.

[0025] In some embodiments, the probe pin **300** includes a pin stopper (also referred to as a pin fin, not shown). In some embodiments, a width W of the pin stopper is greater than a diameter D of the probe pin **300**. Further, a sum of the width W of the pin stopper and the diameter D of the probe pin **300** is greater than the widths of the through holes **214** of the upper die **210**. The pin stopper may be coated on the probe pin **300**, and may be formed of a conductive or a dielectric material. In some embodiments, an entirety of the probe pin **300** has a uniform diameter D . In some alternative embodiments, different portions of the probe pin **300** may have different diameters, though not shown.

[0026] Referring to FIGS. **2** and **3**, in some embodiments, the probe pin **300** has two ends, wherein one end may be referred to as a pin head **302**, and the opposite end may be referred to as a pin tip **304**. When assembling the probe pin **300** into the probe head **200**, the probe pin **300** is inserted into the probe head **200** such that the pin head **302** is exposed through the upper die **210**, while the pin tip **304** is exposed through the lower die **220**. The pin stopper may be disposed within the cavity **C**. As mentioned above, because the sum of the width W of the pin stopper and the diameter D of the probe pin **300** is greater than the width of the through holes **214** of the upper die **210**, the pin stopper helps prevent the probe pin **300** from accidentally falling out of the probe head **200** through the through holes **214** of the upper die **210**. Further, the pin head **302** and the pin tip **304** are aligned with each other; thus, a straight line may be formed as depicted by the dotted line in FIG. **3**.

[0027] In some embodiments, when the probe pin **300** is in contact with a probe pad of a DUT (both not shown), the probe pin **300** may be pushed back by the contact pads, and deformation may occur. To control the deformation of the probe pin **300**, the upper die **210** may be shifted laterally to bend the probe pin **300**. As shown in FIGS. **4A** and **4B**, directions of the lateral shift are indicated by an arrow **D1**. The bending of the probe pin **300** may help prevent the probe pin **300** from shorting with each other. Consequently, positioned of the pin head **302** and the pin tip **304** are offset from each other, and a bending delta Δ may be formed, as shown in FIGS. **4A** and **4B**.

[0028] When the probe card or the probe head **200** reaches its service life end, the probe head **200** is scrapped. However, the probe pin **300** may be re-used. In some embodiments, the bending of the probe pin **300** may cause difficulty in retrieval the probe pin **300**.

[0029] Please refer to FIG. **1**, which is a flowchart representing a method for a probe head pin retrieval **10** according to aspects of the present disclosure. The method **10** includes a number of operations (**11**, **12**, **13**, **14** and **15**). The method **10** will be further described according to one or more embodiments. It should be noted that the operations of the method **10** may be rearranged or otherwise modified within the scope of the various aspects. It should be further noted that additional processes may be provided before, during, and after the method **10**, and that some other processes may be only briefly described herein. Thus, other implementations are possible within the scope of the various aspects described herein.

[0030] FIGS. **5** to **11**, which are schematic drawings illustrating a method for a probe pin retrieval at various stages according to aspects of the present disclosure in one or more embodiments. In

some embodiments, a system for retrieving a probe pin **400** is provided. As shown in FIG. 5, the system **400** includes a carrier **410**, a bending fixture **420**, a plate **430** and an arm **440**. The carrier **410** is configured to accommodate a probe head of a probe card. For example, the carrier **410** can accommodate a probe head **200** as mentioned above. The bending fixture **420** is disposed over the carrier **410** and configured to clamp the upper die **210** of the probe head **200**. Further, the bending fixture **420** moves in a horizontal direction. The plate **430** is disposed under the carrier **410** and configured to push the probe pin **300** of the probe head **200**. The arm **440** is disposed over the carrier **410** and configured to pick the probe pin **300** out of the probe head **200**.

[0031] Referring to FIG. 5, in some embodiments, in operation **11**, the probe head **200** is received in the carrier **410**. As mentioned above, the probe head **200** includes the upper die **210**, the lower die **220**, the spacer **230**, and at least the probe pin **300**. Further, the probe pin **300** extends in a direction from the lower die **220** to the upper die **210**. In some embodiments, the carrier **410** may include a hollow configuration, such that the spacer **230** is disposed on and in contact with the carrier **410**, leaving the lower die **220** exposed. In some embodiments, the carrier **410** may further include a securing element to clamp or secure the probe head **200**, though not shown.

[0032] In some embodiments, in operation **12**, a bending delta Δ is measured. As mentioned above, the bending delta Δ is measured as a lateral distance between the pin head **302** and the pin tip **304** of the probe pin **300**. In some embodiments, a charge coupled device (CCD) may be used to detect and measure the bending delta Δ between the pin head **302** and the pin tip **304**. In some embodiments, when the bending delta Δ is less than a value, it is determined that the retrieval of the probe pin **300** may be performed. In some alternative embodiments, when the bending delta Δ is greater than a value, it is determined that a deformation correction of the probe pin **300** may be required. In some embodiment, the system **400** may include a controller (not shown), and the determination of the retrieval or the deformation correction of the probe pin **300** may be performed by the controller. In other words, the retrieval or the deformation correction of the probe pin **300** may be performed automatically. In some embodiments, the controller can be a central processing unit (CPU), but the disclosure is not limited thereto.

[0033] Additionally, in some embodiments, the operation **12** may be performed prior to the operation **11**.

[0034] Referring to FIG. 6, in some embodiments, in operation **13**, the probe pin **300** is bent by the bending fixture **420** when the bending delta Δ is greater than the value. In some embodiments, the value may be approximately 120 micrometers. That is, when the bending delta Δ is greater than approximately 120 micrometers, a deformation correction is required. However, those skilled in the art can realize that the value may be smaller than 120 micrometers as pitches of probe pads are continuously reduced. As shown in FIG. 6, in some embodiments, the bending fixture **420** may be lowered to clamp the upper die **210**. Subsequently, the bending fixture **420** moves in a direction D2 parallel to a surface of the upper die **210**. In some embodiments, the bending fixture **420** moves in the direction D2 which is parallel to the bending direction D1. Further, the bending fixture **420** moves in the direction D2 opposite to the bending direction D1. Because the upper die **210** is clamped by the bending fixture **420**, the upper die **210** is moved along with the bending fixture **420**, while the lower die **220** and the spacer **230** are secured on the carrier **410**.

[0035] Referring to FIG. 7, in some embodiments, the bending fixture **420** bends the probe pin **300** during the moving of the upper die **210**. Another bending delta Δ' may be obtained after the bending of the probe pin **300**. As mentioned above, the bending delta Δ is defined as a lateral distance between the pin head **304** and the pin tip **306**. In some embodiments, the moving of the bending fixture **420** moves the upper die **210** and bends the probe pin **300** such that the deformation of the probe pin **300** may be mitigated. For example, the bending delta Δ' of the probe pin **300** after the moving of the bending fixture **420** (and the bending of the probe pin **300**) is less than the bending delta Δ before the moving of the bending fixture **420** (and the bending of the probe pin **300**). Accordingly, the bending delta Δ' is less than the value. For example, the bending

delta Δ' is less than 120 micrometers, but the disclosure is not limited thereto.

[0036] Referring to FIG. 8, in operation 14, the probe pin 300 is pushed in a direction from the lower die 220 toward the upper die 210 by the plate 430. In some embodiments, the bending fixture 420 is moved away from the probe head 200, as shown in FIG. 8. Subsequently, the plate 430 is moved toward the lower die 220. The plate 430 is raised to in contact with pin tip 306 of the probe pin 300. Further, the plate 430 pushes the probe 300 in the direction from the lower die 220 toward the upper die 210. Consequently, the probe pin 300 is raised. In some embodiments, a force of the plate 430 is sufficient to push the pin stopper out of the upper die 210, as shown in FIG. 8. As mentioned above, the width of the through hole 214 of the upper die 210 is less than the sum of the diameter of the probe pin 300 and the width of the pin stopper. Therefore, the force of the plate 430 must be great enough to push the pin stopper above a surface of the upper die 210. In other words, the pin stopper is raised to be higher than the surface of the upper die 210.

[0037] In some embodiments, a distance S may be defined between the pin head 302 and the surface of the upper die 210 after the pushing of the probe pin 300. In some embodiments, the distance S is defined between a top surface of the pin head 302 and the surface of the upper die 210. In some embodiments, the distance S may be greater than approximately 200 micrometers, but the disclosure is not limited thereto. In some embodiments, the distance S is made greater than 200 micrometers for a subsequent operation. However, the distance S may be greater than or less than 200 micrometers, according to different system requirements.

[0038] Referring to FIG. 9, in operation 15, the probe pin 300 is picked from the probe head 200 by the arm 440. As mentioned above, because the bending delta Δ' is less than the value, it is easier to pick the probe pin 300. Thus, the difficulty of the retrieval of the probe pin 300 is mitigated.

[0039] Additionally, in some embodiments, when the bending delta Δ of the probe pin 300 is less than the above-mentioned value, the operation 13 may be omitted, and the operation is performed to push the probe pin 300 by the plate 430. Because the bending delta Δ is less than the above-mentioned value, the probe pin 300 may be easily pushed to obtain the distance S between the pin head 304 and the surface of the upper die 210. The operation 15 may be performed to pick the probe pin 300 out of the probe head 200 using the arm 440.

[0040] In some embodiments, the probe pin 300 may be inspected after the retrieval from the probe head 200. Accordingly, a good probe pin 300 may be determined to be re-usable.

[0041] In some embodiments, the probe pin 300 is put in a pin box by the arm 440. The probe pin 300 may be stored and protected in the pin box.

[0042] In some embodiments, the probe pin 300 may be re-used in a new probe head. In such embodiments, the arm 440 may move to clamp and pick the probe pin 300. In some embodiments, operations for the deformation correction may be performed. In some embodiments, operations for straightening the probe pin 300 may be performed. Consequently, a bending delta Δ'' of the probe pin 300 may be obtained. In some embodiments, the bending delta Δ'' of the probe pin 300 is less than the value. For example, the bending delta Δ'' of the probe pin 300 may be less than approximately 120 micrometers, but the disclosure is not limited thereto. In some embodiments, the bending delta Δ'' of the probe pin 300 is equal to or less than the bending delta Δ' and the bending delta Δ . In some embodiments, CCD may be used to measure the bending delta Δ'' .

[0043] Referring to FIG. 11, in some embodiments, a new probe head 500 may be received. The probe head 500 may include upper die 510, lower die 520, and a spacer 530 between the upper and lower dies 510- and 520, but the disclosure is not limited thereto. The upper die 510 may have a plurality of through holes 514. Parameters of the through holes 514 may be similar to those described above; therefore, repeated description is omitted for brevity. The lower die 520 may have a plurality of through holes 524. Parameters of the through holes 524 may be similar to those described above; therefore, repeated description is omitted for brevity. As mentioned above, the through hole 514 of the upper die 510 and the through hole 524 of the lower die 520 are vertically aligned with a one-to-one correspondence.

[0044] In some embodiments, the arm **420** moves the probe pin **300** and assembles the probe pin **300** into the probe head **500**, as shown in FIG. **11**. Because the bending delta Δ'' of the probe pin **300** is equal to or less than the bending delta Δ' , the probe pin **300** can be easily inserted into the probe head **500**. Thus, re-use rate of the probe pin **300** may be increased.

[0045] FIG. **12** is a flowchart representing a method for retrieving a probe pin **50** according to aspects of the present disclosure. The method **50** includes a number of operations (**51**, **52**, **53**, **54**, **55**, **56** and **57**). The method **50** will be further described according to one or more embodiments. It should be noted that the operations of the method **10** may be rearranged or otherwise modified within the scope of the various aspects. It should be further noted that additional processes may be provided before, during, and after the method **50**, and that some other processes may be only briefly described herein. Thus, other implementations are possible within the scope of the various aspects described herein.

[0046] In some embodiments, in operation **51**, a probe head **200** is received. As mentioned above, the probe head **200** includes one or more upper die **210**, one or more lower die **220**, a spacer **230** between the upper and lower dies **210** and **220**, and at least a probe pin **300**. As mentioned above, the upper die **210**, the lower die **220**, and the spacer **230** together form a cavity C. The probe pin **300** passes through the lower die **220**, the cavity C and the upper die **210** through holes **214** and **224**. In some embodiments, the probe pin **300** extends in a direction from the lower die **220** toward the upper die **210**. Further, a pin head **302** of the probe pin **300** protrudes from an outer surface of the upper die **210**, while a pin tip **304** of the probe pin **300** protrudes from an outer surface of the lower die **220**. In some embodiments, the probe pin **300** may include a pin stopper for preventing the probe pin **300** from accidentally falling out of the probe head **200** through the upper die **210**.

[0047] In some embodiments, when the probe pin **300** is in contact with a probe pad of a DUT (both not shown), the probe pin **300** may be pushed back by the contact pads, and deformation may occur. To control the deformation of the probe pin **300**, the upper die **210** may be shifted laterally to bend the probe pin **300**. As shown in FIGS. **4A** and **4B**, the lateral shift directions are indicated by an arrow **D1**. The bending of the probe pins **300** may help prevent the probe pins **300** from shorting with each other. Consequently, positions of the pin head **304** and the pin tip **306** are offset, and a bending delta Δ may be formed between the pin head **304** of the probe pin **300** and the pin tip **306** of the probe pin **300**, as shown in FIGS. **4A** and **4B**. As mentioned above, the bending delta Δ is measured as a lateral distance between the pin head **304** and the pin tip **306** of the probe pin **300**.

[0048] In operation **52**, the bending delta Δ of the probe pin **300** is measured. As mentioned above, a CCD may be used to measure the bending delta Δ of the probe pin **300**. In operation **53**, the bending delta Δ is compared to a value. In some embodiments, the value may be approximately 120 micrometers. That is, when the bending delta Δ is greater than approximately 120 micrometers, a deformation correction is required. However, those skilled in the art can realize that the value may be smaller than 120 micrometers as pitches of probe pads are continuously reduced.

[0049] In operation **54**, it is determined whether the bending delta Δ is greater than the value. In some embodiments, when the bending delta Δ is greater than the value, a positive result is determined, and operation **55** is performed. In some embodiments, the recognizing and the determination may be performed by a controller, as mentioned above.

[0050] Referring to FIG. **7**, in some embodiments, the bending fixture **420** is used to clamp the upper die **210**. The bending fixture **420** moves in a direction **D2** parallel to a surface of the upper die **210**. In some embodiments, the bending fixture **420** moves in the direction **D2**, which is parallel to the bending direction **D1**. Further, the bending fixture **420** moves in the direction **D2** opposite to the bending direction **D1**. Because the upper die **210** is clamped by the bending fixture **420**, the upper die **210** is moved along with the bending fixture **420**, while the lower die **220** and the spacer **230** are secured on the carrier **410**. In some embodiments, the moving of the bending fixture **420** moves the upper die **210** and bends the probe pin **300** such that the deformation of the probe pin **300** may be mitigated.

[0051] Referring to FIG. 8, in operation 56, the probe pin 300 is pushed in a direction from the lower die 220 toward the upper die 210 by the plate 430. In some embodiments, the bending fixture 420 is moved away from the probe head 200, as shown in FIG. 8. Subsequently, the plate 430 is moved toward to the lower die 220. The plate 430 is raised to contact the pin tip 304 of the probe pin 300. Further, the plate 430 pushes the probe pin 300 in the direction from the lower dies 220 toward the upper die 210. Consequently, the probe pin 300 is raised, as shown in FIG. 8. In some embodiments, a force of the plate 430 is sufficient to push the pin stopper out of the upper die 210. As mentioned above, the width of the through hole 214 of the upper die 210 is less than the sum of the diameter of the probe pin 300 and the width of the pin stopper. Therefore, the force of the plate 430 must be great enough to push the pin stopper above the surface of the upper die 210. In other words, the pin stopper is raised to be higher than the surface of the upper die 210.

[0052] As mentioned above, a distance S may be defined between the pin head 302 and the surface of the upper die 210 after the pushing of the probe pin 300. In some embodiments, the distance S is defined between a top surface of the pin head 302 and the surface of the upper die 210. In some embodiments, the distance S may be greater than approximately 200 micrometers, but the disclosure is not limited thereto. In some embodiments, the distance S is made greater than 200 micrometers for a subsequent operation. However, the distance S may be greater than or less than 200 micrometers, according to different system requirements.

[0053] In some embodiments, in operation 57, the probe pin 300 is picked from the probe head 200 by an arm 440. As mentioned above, because the bending delta Δ' is less than the value, it is easier to pick the probe pin 300. Thus, the difficulty of the retrieval of the probe pin 300 is mitigated.

[0054] Alternatively, in some embodiments, when the bending delta Δ is less than the value, a negative result is determined in operation 54. In such embodiments, operation 55 is not performed, and operations 56 and 57 are performed.

[0055] According to one embodiment of the present disclosure, a method for retrieving a probe pin are provided. The method uses a bending fixture of the system to move an upper die of a probe head, thereby bending the probe pin. Consequently, a bending delta of the probe pin is reduced such that the probe pin can be easily retrieved from the probe head. The retrieved probe pin may be stored in a pin box, and may be further treated to mitigate a deformation issue. Such probe pin may be re-used in a new probe head. Accordingly, a re-use rate of the probe pin is improved.

[0056] According to one embodiment of the present disclosure, a method for retrieving a probe pin is provided. The method includes following operations. A probe head is received in a carrier. The probe head includes an upper die, a lower die, and at least a probe pin extending in a direction from the lower die to the upper die. A first bending delta between a pin tip of the probe pin and a pin head of the probe pin is measured. The probe pin is bent by a bending fixture when the first bending delta is greater than a value to obtain a second bending delta between the pin tip and the pin head. The probe pin is pushed in the direction from the lower die to the upper die by a plate. The probe pin is picked from the probe head by an arm.

[0057] According to one embodiment of the present disclosure, a method for retrieving a probe pin is provided. The method includes following operations. A probe head is received in a carrier. The probe head includes an upper die, a lower die, and at least a probe pin extending in a direction from the lower die to the upper die. A bending fixture is used to clamp the upper die and the move the upper die so as to bend the probe pin in a direction parallel to a surface of the upper die. The probe pin has a first bending delta prior to the moving of the upper die and a second bending delta after the moving of the upper die. The second bending delta is less than the first bending delta.

[0058] According to one embodiment of the present disclosure, a method for retrieving a probe pin is provided. The method includes following operations. A probe head is received in a carrier. The probe head includes an upper die, a lower die, and at least a probe pin extending in a direction from the lower die to the upper die. A first bending delta between a pin tip of the probe pin and a pin head of the probe pin is measured using a charge-coupled device. A deformation correction is

performed when the first bending delta is greater than a value. The deformation correction further includes using a bending fixture to clamp the upper die of the probe head, and moving the bending fixture to bend the probe pin to obtain a second bending delta between the pin tip and the pin head. The probe pin is pushed in the direction from the lower die to the upper die by a plate after the performing of the deformation correction or when the first bending delta is less than the value. The probe pin is picked from the probe head by an arm.

[0059] According to one embodiment of the present disclosure, a method for re-using a probe pin is provided. The method includes following operations. A first probe head including at least a probe pin is received. A first bending delta between a pin tip of the probe pin and a pin head of the probe pin is measured. The first bending delta is compared with a value. A recognition if the first bending delta is greater than the value or not is performed. The probe pin is bended to obtain a second bending delta when the first bending delta is greater than the value. The probe pin is retrieving from the first probe head. The probe pin is assembled to a second probe head. The probe pin in the second probe head has a third bending delta between the pin tip and the pin head. The second bending delta and the third bending delta are less than the first bending delta.

[0060] According to one embodiment of the present disclosure, a method for retrieving a probe pin is provided. The method includes following operations. A first probe head including an upper die, a lower die, and at least a probe pin is received. A first bending delta between a pin tip of the probe pin and a pin head of the probe pin is obtained. The first bending delta is compared with a value. A recognition is performed to find if the first bending delta is less than the value or not. The probe pin is pushed in a direction from the lower die to the upper die when the first bending delta is less than the value. The probe pin is assembled to a second probe head. The probe pin in the second probe head has a second bending delta between the pin tin and the tin head. The second bending delta is less than the first bending delta.

[0061] According to one embodiment of the present disclosure, a method for re-using a probe pin is provided. The method includes following operations. A first probe head including an upper die, a lower die, and at least a probe pin is received. A first bending delta between a pin tip of the probe pin and a pin head of the probe pin is compared with a value. The probe pin is bended to obtain a second bending delta between the pin tip and the pin head when the first bending delta is greater than the value. The probe pin is pushed in a first direction from the lower die to the upper die after the bending of the probe pin, or when the first bending delta is less than the value. The probe pin is retrieved from the first probe head. The probe pin is assembled to a second probe head. A third bending delta between the pin tip and the pin head of the probe pin in the second probe head is less than the first bending delta.

[0062] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method for re-using a probe pin, comprising: receiving a first probe head comprising at least a probe pin; measuring a first bending delta between a pin tip of the probe pin and a pin head of the probe pin; comparing the first bending delta with a value; recognizing if the first bending delta is greater than the value or not; bending the probe pin when the first bending delta is greater than the value to obtain a second bending delta; retrieving the probe pin from the first probe head; and

assembling the probe pin into a second probe head, wherein the probe pin in the second probe head has a third bending delta between the pin tip and the pin head, wherein the second bending delta and the third bending delta are less than the first bending delta.

2. The method of claim 1, wherein the second bending delta and the third bending delta are less than the value.

3. The method of claim 1, wherein the first probe head comprises an upper die and a lower die, and the probe pin extends in a first direction from the lower die to the upper die.

4. The method of claim 3, wherein the bending of the probe pin further comprises: clamping the upper die of the first probe head by a bending fixture; and moving the bending fixture to bend the probe pin in a second direction different from the first direction.

5. The method of claim 1, wherein the measuring of the first bending delta is performed by a charge-coupled device (CCD).

6. The method of claim 1, wherein the retrieving of the probe pin from the first probe head is performed by an arm.

7. The method of claim 6, wherein the assembling of the probe pin into the second probe head is performed by the arm.

8. A method for retrieving a probe pin, comprising: receiving a first probe head comprising an upper die, a lower die, and at least a probe pin; obtaining a first bending delta between a pin tip of the probe pin and a pin head of the probe pin; comparing the first bending delta with a value; recognizing if the first bending delta is less than the value or not; pushing the probe pin in a direction from the lower die to the upper die when the first bending delta is less than the value; and assembling the probe pin into a second probe head, wherein the probe pin in the second probe head has a second bending delta between the pin tip and the pin head, wherein the second bending delta is less than the first bending delta.

9. The method of claim 8, wherein the value is greater than approximately 120 micrometers.

10. The method of claim 8, wherein the second bending delta is less than approximately 120 micrometers.

11. The method of claim 8, wherein the upper die of the first probe head comprises a through hole allowing the probe pin through.

12. The method of claim 8, further comprising: using a plate to push the probe pin in the direction from the lower die to the upper die of the first probe head; and using an arm to pick the pin and assemble the probe pin to the second probe head.

13. The method of claim 8, further comprising using a charge-coupled device (CCD) to measure the first bending delta and the second bending delta.

14. A method for re-using a probe pin, comprising: receiving a first probe head comprising an upper die, a lower die, and at least a probe pin; comparing a first bending delta between a pin tip of the probe pin and a pin head of the probe pin with a value; bending the probe pin to obtain a second bending delta between the pin tip and the pin head when the first bending delta is greater than the value; pushing the probe pin in a first direction from the lower die to the upper die after the bending of the probe pin, or when the first bending delta is less than the value; retrieving the probe pin from the first probe head; and assembling the probe pin into a second probe head, wherein a third bending delta between the pin tip of the probe pin and the pin head of the probe pin in the second probe head is less than the first bending delta.

15. The method of claim 14, wherein the third bending delta is equal to or less than the second bending delta.

16. The method of claim 14, wherein the second bending delta is less than the first bending delta.

17. The method of claim 14, further comprising straightening the probe pin prior to the assembling of the probe pin into the second probe head.

18. The method of claim 17, wherein the straightening of the probe pin obtains the third bending data.

19. The method of claim 14, wherein the bending of the probe pin further comprises: clamping the upper die of the first probe head by a bending fixture; and moving the bending fixture in a second direction different from the first direction.

20. The method of claim 14, further comprising using a plate under the lower die to push the probe pin in the first direction.
